usepackage[a4paper,margin=0.65in]geometry

Hence if the function $\zeta = \dfrac{(a-b)(b-c)}{(a-b)(c-d)}$ satisfy this condition, the indefinite integral

$$\int \frac{(x-a)^\alpha (x-b)^\beta \, dx}{(x-c)^\gamma (x-d)^\delta},$$

where $\gamma+\delta=\alpha+\beta+2$, will be expressible as a rational algebraical fraction.

It may be noticed, that in the general case, observing that $x=a$, $x=b$ give $y=0$, $y=1$, the integral

$$\int_a^b (x-a)^\alpha (x-b)^\beta (x-c)^\gamma (x-d)^\delta \, dx$$

depends on

$$\int_0^1 y^\alpha (1-y)^\beta (\zeta-y)^\delta \, dy,$$

or, putting $\zeta = \dfrac{1}{u}$, upon

$$\int_0^1 y^\alpha (1-y)^\beta (1-uy)^\delta \, dy,$$

which is expressible by means of a hypergeometric series having u for its argument or fourth element.

2, Stone Buildings, Lincoln's Inn, Feb. 1854.

160.

On a Theorem Relating to Reciprocal Triangles.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 7–10.]

The following theorem is, I assume, known; but the analytical demonstration of it depends upon a formula in determinants which is not without interest. The theorem referred to may be thus stated:

"A triangle and its reciprocal are in perspective," where by the reciprocal of a triangle is meant the triangle the sides of which are the polars of the angles of the first-mentioned triangle with respect to a conic; and triangles are in perspective when the three lines forming the corresponding angles meet in a point, or what is the same thing, when the three points of intersection of the corresponding sides lie in a line.

Let the equation of the conic be

$$x^2 + y^2 + z^2 = 0,$$

and take (α, β, γ), $(\alpha', \beta', \gamma')$, $(\alpha'', \beta'', \gamma'')$ for the coordinates of the angles of the triangle, then if K be the determinant, and (A, B, C) (A', B', C') (A'', B'', C'') the inverse system, i.e. if

$$\begin{aligned}
KA &= (\beta'\gamma'' - \beta''\gamma'), & KB &= \gamma'\alpha'' - \gamma''\alpha', & KC &= \alpha'\beta'' - \alpha''\beta', \\
KA' &= (\beta''\gamma - \beta\gamma''), & KB' &= \gamma''\alpha - \gamma\alpha'', & KC' &= \alpha''\beta - \alpha\beta'', \\
KA'' &= (\beta\gamma' - \beta'\gamma), & KB'' &= \gamma\alpha' - \gamma'\alpha, & KC'' &= \alpha\beta' - \alpha'\beta,
\end{aligned}$$

equations which may be represented in the notation of matrices by the single equation

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix}^{-1} = \begin{vmatrix} A, & A', & A'' \\ B, & B', & B'' \\ C, & C', & C'' \end{vmatrix},$$

then the equations of the sides of the triangle are

$$\begin{aligned} A\,x + B\,y + C\,z &= 0,\\ A'\,x + B'\,y + C'\,z &= 0,\\ A''\,x + B''\,y + C''\,z &= 0,\end{aligned}$$

and the coordinates of the angles of the reciprocal triangle may be taken to be (A, B, C) (A', B', C') (A'', B'', C''); the equations of the lines joining the corresponding angles of the two triangles are therefore

$$\begin{aligned} (B\gamma - C\beta)\,x + (C\alpha - A\gamma)\,y + (A\beta - B\alpha)\,z &= 0,\\ (B'\gamma' - C'\beta')\,x + (C'\alpha' - A'\gamma')\,y + (A'\beta' - B'\alpha')\,z &= 0,\\ (B''\gamma'' - C''\beta'')\,x + (C''\alpha'' - A''\gamma'')\,y + (A''\beta'' - B''\alpha'')\,z &= 0;\end{aligned}$$

the condition that these lines may meet in a point is therefore

$$\begin{vmatrix} B\gamma - C\beta, & C\alpha - A\gamma, & A\beta - B\alpha \\ B'\gamma' - C'\beta', & C'\alpha' - A'\gamma', & A'\beta' - B'\alpha' \\ B''\gamma'' - C''\beta'', & C''\alpha'' - A''\gamma'', & A''\beta'' - B''\alpha'' \end{vmatrix} = 0,$$

an equation which is satisfied identically when A, B, C; A', B', C'; A'', B'', C'' are replaced by their values. To prove this I transform the different quantities which enter into the determinant as follows: putting

$$\begin{aligned} F &= \alpha'\alpha'' + \beta'\beta'' + \gamma'\gamma'',\\ G &= \alpha''\alpha + \beta''\beta + \gamma''\gamma,\\ H &= \alpha\alpha' + \beta\beta' + \gamma\gamma';\end{aligned}$$

we have

$$\begin{aligned} K(B\gamma - C\beta) &= \gamma\,(\gamma''\alpha' - \gamma'\alpha'') - \beta\,(\alpha'\beta'' - \alpha''\beta')\\ &= \alpha''(\beta\beta' + \gamma\gamma') - \alpha'(\beta\beta'' + \gamma\gamma'')\\ &= \alpha''(\alpha\alpha' + \beta\beta' + \gamma\gamma') - \alpha'(\alpha\alpha'' + \beta\beta'' + \gamma\gamma'')\\ &= \alpha''H - \alpha'G,\\ \&\text{c.};\end{aligned}$$

and the equation becomes

$$\begin{vmatrix} \alpha''H - \alpha'G, & \beta''H - \beta'G, & \gamma''H - \gamma'G \\ \alpha F - \alpha''H, & \beta F - \beta''H, & \gamma F - \gamma''H \\ \alpha'G - \alpha F, & \beta'G - \beta F, & \gamma'G - \gamma F \end{vmatrix} = 0.$$

Now the minor $(\beta F - \beta'' H)(\gamma' G - \gamma F) - (\gamma F - \gamma'' H)(\beta' G - \beta F)$ is equal to

$$GH(\beta'\gamma'' - \beta''\gamma') + HF(\beta''\gamma - \beta\gamma'') + FG(\beta\gamma' - \beta'\gamma),$$

i.e. to

$$K\,(GHA + HFA' + FGA'');$$

and expressing the other minors in a similar form, the equation to be proved is

$$\begin{aligned}&(GHA + HFA' + FGA'')(B\gamma - C\beta)\\ +&(GHB + HFB' + FGB'')(C\alpha - A\gamma) = 0,\\ +&(GHC + HFC' + FGC'')(A\beta - B\alpha)\end{aligned}$$

i.e.

$$HF\begin{vmatrix} A', & B', & C' \\ A, & B, & C \\ \alpha, & \beta, & \gamma \end{vmatrix} + FG\begin{vmatrix} A'', & B'', & C'' \\ A, & B, & C \\ \alpha, & \beta, & \gamma \end{vmatrix} = 0.$$

The first determinant is

$$-\{\alpha(B'C-BC')+\beta(C'A-CA')+\gamma(A'B-AB')\} = -\frac{1}{K}(\alpha\alpha''+\beta\beta''+\gamma\gamma'') = -\frac{1}{K}G,$$

and the second determinant is

$$\{\alpha(B''C-BC'')+\beta(C''A-CA'')+\gamma(A''B-AB'')\} = \frac{1}{K}(\alpha\alpha'+\beta\beta'+\gamma\gamma') = \frac{1}{K}H,$$

and we have therefore identically

$$HF\,(-G) + FG\,(H) = 0.$$

The corresponding theorem in geometry of three dimensions is that a tetrahedron and its reciprocal have to each other a certain relation, viz. the four lines joining the corresponding angles are generating lines of a hyperboloid, or, what is the same thing, the four lines of intersection of corresponding faces are generating lines of a hyperboloid. The demonstration would show how the theorem in determinants is to be generalised.

2, Stone Buildings, Lincoln's Inn, February, 1855.

161.

A Problem in Permutations.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), p. 79.]

The game called Mousetrap gives rise to a singular problem in permutations. A set of cards, ace, two, three, &c., say up to thirteen, are arranged in a circle with their faces upwards—you begin at any card, and count one, two, three, &c., and if upon counting suppose the number five, you arrive at the card five, that card is thrown out; and beginning again with the next card, you count one, two, three, &c., throwing out if the case happen a new card as before, and so on until you have counted up to thirteen, without coming to a card which ought to be thrown out. It is easy to see that, whatever the number of the cards is, they may be so arranged as to be all thrown out in the order of their numbers; but that it is not possible in general to arrange the cards so that all the cards, or any specified cards, may be thrown out in a given order. Thus, if all the cards are to be thrown out in the order of their numbers, the arrangements in the case of a single card, two, three &c. cards, are

1							
1	2						
1	3	2					
1	4	2	3				
1	3	2	5	4			
1	4	2	5	6	3		
1	5	2	7	4	3	6	
1	6	2	4	5	3	7	8

&c.

It is required to investigate the general theory.

162.

Two Letters on Cubic Forms.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 85–87 and 90–91.]

Cher Mons. Hermite,

Il y a longtemps que j'ai voulu vous écrire, mais je n'ai été empêché je ne sais comment; j'ai assez à vous dire par rapport aux covariants, mais à présent je vais vous parler des formes cubiques à deux indéterminées. Il me semble que l'on peut simplifier la théorie de Eisenstein, et l'étendre au cas d'un déterminant négatif quelconque, de la manière que voici.

Soit $(a, b, c, d \between x, y)^3$ une forme cubique, je représente par Hess. $(a, b, c, d \between x, y)^2$ la forme quadratique dérivée $(ac - b^2,\ \frac{1}{2}(ad - bc),\ bd - c^2 \between x, y)^2$. Cela étant, soit (A, B, C) une forme représentative (réduite et proprement primitive) au déterminant $-D$; à moins que $(A, B, C)^p = (A, -B, C)$, c'est-à-dire, à moins qu'on n'ait, ou que la nouvelle forme laquelle par sa triplication produit la forme principale, il n'existe pas de forme cubique (a, b, c, d) telle que $b^2 - ac = A$, $bc - ad = 2B$, $c^2 - bd = C$; mais en supposant que l'on ait $(A, B, C)^p = (A, -B, C)$ on peut trouver une seule forme cubique qui satisfait à l'équation dont il s'agit. J'écarte, cela va sans dire, l'une ou l'autre des deux formes (a, b, c, d) et $(-a, -b, -c, -d)$.

En effet on a identiquement

$$\begin{aligned}&(b^2 - ac,\ \tfrac{1}{2}(bc - ad),\ c^2 - bd \between bx + cy,\ cx + dy)^2\\ &= (b^2 - ac,\ \tfrac{1}{2}(bc - ad),\ c^2 - bd \between y, (b^2 - ac,\ \tfrac{1}{2}(bc - ad),\ c^2 - bd \between x',\ y')^2,\end{aligned}$$

donc, en supposant que $b^2 - ac = A$, $bc - ad = 2B$, $c^2 - bd = C$, il s'ensuit que $(A, B, C)^p = (A, -B, C)$.

Je suppose donc $(A, B, C)^p = (A, -B, C)$, et je dis qu'il ne peut pas y avoir deux formes cubiques, (a, b, c, d) et (a_1, b_1, c_1, d_1), qui aient la propriété dont il s'agit; car, en écrivant

$$\left.\begin{aligned}\xi &= bxx' + cxy' + cx'y + dyy',\\ \eta &= axx' + bxy' + bx'y + cyy',\end{aligned}\right\} \qquad \left.\begin{aligned}\xi_1 &= b_1xx' + c_1xy' + c_1x'y + d_1yy',\\ \eta_1 &= a_1xx' + b_1xy' + b_1x'y + c_1yy',\end{aligned}\right\}$$

on trouverait

$$(A, B, C \between \xi, \eta)^2 = (A, -B, C \between \xi_1, \eta_1)^2,$$

ce qui implique d'abord que ξ_1, η_1 soient des fonctions linéaires de ξ, η. Mais $(A, -B, C)$ étant une forme réduite et proprement primitive au déterminant $-D$, il n'existe pas de transformation de la forme quadratique en elle-même, hormis $\xi = \xi$, $\eta = \eta$. Le cas $D = 1$ doit se traiter à part; dans ce cas particulier il n'y a que la forme cubique $(0, 1, 0, 1)$. Donc &c. Enfin, si $(A, B, C)^p = (A, -B, C)$, il existe une forme cubique (a, b, c, d) telle que $b^2 - ac = A$, &c.; car en cherchant par la méthode de Gauss les valeurs des coefficients p, p', p'', p''' et q, q', q'', q''' qui donnent cette transformation, on obtient l'identité $p^2 = aq$, $pp' = bq$, $p'p'' = cq$, $p''^2 = dq$. On peut donc représenter ces coefficients par b_1, c_1, c_1, d_1; a_1, b_1, b_1, c_1; savoir, en peut trouver a, b, c, d, de manière que

$$\begin{aligned}(A, -B, C \between bxx' + cxy' + cx'y + dyy',\ axx' + bxy' + bx'y + cyy')^2&\\ = (A, B, C \between x, y) \cdot (A, B, C \between x', y').&\end{aligned}$$

Cela étant, les équations de Gauss donnent

$$\begin{aligned}A &= bb_1 - ac_1, & A &= b^2 - ac,\\ 2B &= cc_1 - ad_1, & 2B &= cb_1 + ad_1 - 2bc_1,\\ C &= cc_1 - bd_1, & C &= c_1^2 - b_1d_1,\end{aligned}$$

et de là on obtient

$$\begin{aligned}b\,(b - b_1) - a\,(c - c_1) &= 0,\\ c\,(b - b_1) - b\,(c - c_1) &= 0,\\ c_1(b - b_1) - b_1(c - c_1) &= 0,\\ d_1(b - b_1) - c_1(c - c_1) &= 0,\end{aligned}$$

c'est-à-dire, ou $\dfrac{a}{b} = \dfrac{b}{c} = \dfrac{b_1}{c_1} = \dfrac{c_1}{d_1}$, ce qui n'est pas vrai (car cela donnerait $A = 0$, $B = 0$, $C = 0$), ou $b - b_1 = 0$, $c - c_1 = 0$; donc $b_1 = b$, $c_1 = c$; et en écrivant d au lieu de d_1, on voit que l'équation de transformation devient

$$(A, -B, C \between bxx' + cxy' + cx'y + dyy',\ axx' + bxy' + bx'y + cyy')^2 = (A, B, C \between x, y) \cdot (A, B, C \between x', y'),$$

où

$$A = b^2 - ac, \quad 2B = bc - ad, \quad C = c^2 - bd. \qquad \text{C. } \eta. \text{ f. à d.}$$

Je ne sais pas si je vous ai mentionné que j'ai calculé les formes quadratiques pour les treize numéros -307, &c. aux déterminants irréguliers. Pour $D = -307$ ces formes sont:

Ordre P. P., 1 genre, 9 classes, c'est-à-dire,

$$\left.\begin{array}{ccc} 1, & \delta, & \delta^2 \\ \delta, & \delta\delta, & \delta\delta_1 \\ \delta_1^2, & \delta\delta_1^2, & \delta\delta_1^4 \end{array}\right| \cdot \frac{m}{307} \left| \begin{array}{l} (1,\ 0,\ 307),\ (7,\ 1,\ 44),\ (7,\ -1,\ 44), \\ (4,\ 1,\ 77),\ (11,\ -1,\ 28),\ (17,\ 4,\ 19), \\ (4,\ -1,\ 77),\ (17,\ -4,\ 19),\ (11,\ 1,\ 28), \end{array}\right.$$

où $\delta^3 = 1$, $\delta_1^3 = 1$.

Ordre I. P., 1 genre, 3 classes, c'est-à-dire,

$$\alpha, \quad \alpha\delta, \quad \alpha\delta^2 \ \Big| \cdot \Big| (2, 1, 154),\ (14, 1, 22),\ (14, -1, 22).$$

A chaque forme de l'ordre P. P. correspond donc une forme et une seule forme cubique au déterminant -1228. Ces formes sont

$$\begin{array}{lll} (0, 1,\ 0,\ -307), & (1,\ -6,\ 8), & (1,\ -1,\ -6,\ -8), \\ (0, 2,\ 1,\ -38), & (1,\ -3,\ -2,\ 8), & (4,\ 1,\ -4,\ -3), \\ (0, 2,\ -1,\ -38), & (4,\ -1,\ -4,\ 3), & (1,\ 3,\ -2,\ 8). \end{array}$$

Je serai bien aise d'avoir de vos nouvelles, je vous prie de me croire votre très-dévoué

A. Cayley.

Cher Mons. Hermite,

On démontre sans peine la proposition avancée dans ma dernière lettre, savoir qu'en supposant

$$\begin{aligned} \xi &= bxx' + cxy' + cx'y + dyy', \\ \eta &= axx' + bxy' + bx'y + cyy', \\ \xi_1 &= b_1xx' + c_1xy' + c_1x'y + d_1yy', \\ \eta_1 &= a_1xx' + b_1xy' + b_1x'y + c_1yy', \\ (A, -B, C \between \xi, \eta)^2 &= (A, B, C \between \xi_1, \eta_1)^2, \end{aligned}$$

on doit avoir

$$\begin{aligned} \xi_1 &= \alpha\xi + \beta\eta, \\ \eta_1 &= \gamma\xi + \delta\eta. \end{aligned}$$

2–2

où α, β, γ, δ sont des entiers. En effet on trouve d'abord en éliminant par ex. xy' et $x'y$,

$$\xi_1 = \alpha\xi + \beta\eta + \lambda xx' + \mu yy',$$
$$\eta_1 = \gamma\xi + \delta\eta + \nu xx' + \rho yy',$$

où α, β, &c., λ, μ, &c., sont des quantités rationnelles. En substituant ces valeurs de ξ_1, η_1, on aura $(A, -B, C \between \lambda, \nu)^2 = (A, B, C \between \mu, \rho)^2 = 0$. Donc le déterminant n'étant pas un carré, $\lambda = 0$, $\nu = 0$, $\mu = 0$, $\rho = 0$, et

$$\xi_1 = \alpha\xi + \beta\eta,$$
$$\eta_1 = \gamma\xi + \delta\eta,$$

où α, β, γ, δ sont des quantités rationnelles. Donc

$$\begin{aligned} b_1 &= \alpha b + \beta a, & a_1 &= \gamma b + \delta a, \\ c_1 &= \alpha c + \beta b, & b_1 &= \gamma c + \delta b, \\ d_1 &= \alpha d + \beta c, & c_1 &= \gamma d + \delta c, \end{aligned}$$

donc α, β seront des quantités rationnelles ayant pour dénominateur l'une quelconque à volonté des trois quantités $b^2 - ac$, $c^2 - bd$, $ad - bc$, c'est-à-dire, des quantités A, B, C; et, puisque (A, B, C) est une forme P. P., cela ne peut arriver à moins que α, β ne soient des entiers; de même, γ, δ seront des entiers. Je remarque de plus les équations

$$\alpha\beta + b\,(\alpha - \delta) - c\gamma = 0,$$
$$b\delta + c\,(\alpha - \delta) - d\gamma = 0,$$

lesquelles donnent

$$\beta : \alpha - \delta : -\gamma = bd - c^2 : bc - ad : ac - b^2 = C : -2B : A.$$

cela fait voir que la transformation en elle-même de la forme $(A, -B, C \between \xi, \eta)^2$ à moyen de $\xi_1 = \alpha\xi + \beta\eta$, $\eta_1 = \gamma\xi + \delta\eta$ est une transformation propre. J'aime cependant mieux le raisonnement dont vous vous êtes servi pour déduire d'une forme cubique quadratique. Il toutes les autres formes cubiques qui correspondent à la même forme quadratique. Il serait facile de la même manière, étant donnée une transformation quelconque d'une forme quadratique dans le produit de deux autres formes quadratiques, d'en déduire toutes les autres transformations; car il y a pour la fonction $axx' + \&c.\ldots$ un covariant qui correspond au cubicovariant de la fonction $ax^3 + \&c.\ldots$ Je suis votre très-dévoué

A. Cayley.

2, Stone Buildings, Lincoln's Inn, 6 Mars, 1855.

163.

On Hansen's Lunar Theory.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 112–125.]

The following paper was written in order to exhibit, in as clear a form as may be, the investigation of the remarkable equations for the motion of the moon established in Hansen's "Fundamenta Nova Investigationis Orbitæ veræ quam Luna perlustrat," &c., Gothae, 1838. I have availed myself for this purpose of the remarks in Jacobi's two letters in answer to a letter of Hansen's, *Crelle*, t. XLII. [1851], p. 12; it may be convenient to remark that the quantity there represented by λ, and which does not occur in Hansen's own investigation, is in this paper represented by Θ.

The position of the moon referred to the earth as centre is determined by

r, the radius vector,
L, the longitude,
Λ, the latitude.

Suppose, moreover, that the attractive force at distance unity, $= \kappa\,(M + E)$, is represented by n^2a^3, then the principal function will be $V = -\dfrac{n^2a^3}{r}$, and the disturbing function R may be represented by $\dfrac{n^2a^3}{r}\Omega$; the expression for the half of the vis viva is

$$T = \tfrac{1}{2}\left\{\left(\frac{dr}{dt}\right)^2 + r^2\cos^2\Lambda\left(\frac{dL}{dt}\right)^2 + r^2\left(\frac{d\Lambda}{dt}\right)^2\right\},$$

and the equations of motion are therefore

$$\begin{aligned} \frac{d\,dr}{dt\,dt} - r\cos^2\Lambda\left(\frac{dL}{dt}\right)^2 - r\left(\frac{d\Lambda}{dt}\right)^2 + \frac{n^2a^3}{r^2} &= n^2a^3\,\frac{d\Omega}{dr}\,, \\ \frac{d}{dt}\left(r^2\cos^2\Lambda\,\frac{dL}{dt}\right) &= n^2a^3\,\frac{d\Omega}{dL}\,, \\ \frac{d}{dt}\left(r^2\frac{d\Lambda}{dt}\right) + r^2\cos\Lambda\sin\Lambda\left(\frac{dL}{dt}\right)^2 &= n^2a^3\,\frac{d\Omega}{d\Lambda}\,, \end{aligned}$$

where Ω is considered as a function of r, L, Λ.

Consider the orbit as an ellipse, then putting

a, the mean distance,
n, the mean motion $= \sqrt{\dfrac{\kappa\,(M+E)}{a^3}}$,
e, the eccentricity,
c, the mean anomaly at epoch,
ω, the distance of perigee from node,
θ, the longitude of node,
i, the inclination,
Φ, the distance from node,
Ψ, the reduced distance from node, $= L - \theta$,
U, the eccentric anomaly,
f, the true anomaly, $= \Phi - \omega$,

the elements of the orbit are a, e, c, ω, θ, i, and we have from the theory of elliptic motion,

$$nt + c = U - e\sin U,$$

$$f = \tan^{-1} \frac{\sqrt{(1-e^2)}\sin U}{\cos U - e},$$

$$r = a\,(1 - e\cos U) = \frac{a\,(1-e^2)}{1 + e\cos f}.$$

Moreover i is the angle at the base of a right-angled spherical triangle, the base, perpendicular, and hypothenuse of which are Ψ, Λ, Φ, hence

$$\tan\Psi = \cos i\,\tan\Phi,$$
$$\sin\Lambda = \sin i\,\sin\Phi,$$
$$\sin\Psi = \cot i\,\tan\Lambda,$$
$$\cos\Phi = \cos\Lambda\,\cos\Psi.$$

Considering the elements as constant, we have

$$\begin{aligned}
\frac{dr}{dt} &= \frac{nae\sin f}{\sqrt{(1-e^2)}}\,,\\
\frac{df}{dt} &= \frac{na^2\sqrt{(1-e^2)}}{r^2}\,,\\
\frac{d\Phi}{dt} &= \frac{na^2\sqrt{(1-e^2)}}{r^2}\,,\\
\frac{d\Psi}{dt} &= \frac{\cos i}{\cos^2\Lambda}\,\frac{na^2\sqrt{(1-e^2)}}{r^2}\,,\\
\frac{dL}{dt} &= \frac{\cos i}{\cos^2\Lambda}\,\frac{na^2\sqrt{(1-e^2)}}{r^2}\,,\\
\frac{d\Lambda}{dt} &= \sin i\,\cos\Psi\,\frac{na^2\sqrt{(1-e^2)}}{r^2}\,.
\end{aligned}$$

Hence also

$$
\begin{aligned}
\frac{d}{dt}\left(\frac{dr}{dt}\right) &= \frac{n^2a^4e\cos f}{r^2}\,,\\
\frac{d}{dt}\left(r^2\cos^2\Lambda\,\frac{dL}{dt}\right) &= 0,\\
\frac{d}{dt}\left(r^2\frac{d\Lambda}{dt}\right) &= -\frac{\cos^2 i\,\sin\Lambda}{\cos^2\Lambda}\,\frac{n^2a^4(1-e^2)}{r^2}\,,\\
\frac{d}{dt}\left(\frac{df}{dt}\right) &= -\frac{2na^2e}{r^3}\,\dot{e}\sin f.
\end{aligned}
$$

The foregoing values show that the equations of motion, neglecting the terms which involve the disturbing functions, are satisfied by the elliptic values of r, L, Λ: and in order to satisfy the actual equations of motion, we have only to consider the elements as variable and to write

$$
\begin{aligned}
dr &= 0,\\
dL &= 0,\\
d\Lambda &= 0,\\
d\,\frac{dr}{dt} &= n^2a^3\,\frac{d\Omega}{dr}\,dt,\\
d\left(r^2\cos^2\Lambda\,\frac{dL}{dt}\right) &= n^2a^3\,\frac{d\Omega}{dL}\,dt,\\
d\left(r^2\frac{d\Lambda}{dt}\right) &= n^2a^3\,\frac{d\Omega}{d\Lambda}\,dt,
\end{aligned}
$$

where the differentiations relate only to the elements, or, what is the same thing, to t in so far only as it enters through the variable elements: the system is at once transformed into

$$
\begin{aligned}
dr &= 0,\\
dL &= 0,\\
d\Lambda &= 0,\\
d\,\frac{nae\sin f}{\sqrt{(1-e^2)}} &= n^2a^3\,\frac{d\Omega}{dr}\,dt,\\
d\,na^2\sqrt{(1-e^2)}\,\cos i &= n^2a^3\,\frac{d\Omega}{dL}\,dt,\\
d\,na^2\sqrt{(1-e^2)}\,\sin i\,\cos\Psi &= n^2a^3\,\frac{d\Omega}{d\Lambda}\,dt.
\end{aligned}
$$

Now $\Psi = L - \theta$, (or supposing, as before, that the differentiations relate to t, only in so far as it enters through the variable elements) $d\Psi = -d\theta$, and thence $d\theta = \dfrac{\tan\Psi}{\sin i}\,di$; we have also $d\Phi = -\cos i\,d\theta$. The equations containing $\dfrac{d\Omega}{dL}$ and $\dfrac{d\Omega}{d\Lambda}$ give

$$\cos i\,d\,na^2\sqrt{(1-e^2)} - na^2\sqrt{(1-e^2)}\sin i\,di = n^2a^3\,\frac{d\Omega}{dL}\,dt,$$

$$\cos i\,\sin i\,d\,na^2\sqrt{(1-e^2)} + na^2\sqrt{(1-e^2)}(\cos\Psi\,\cos i\,di + \sin i\,\sin\Psi\,d\theta) = n^2a^3\,\frac{d\Omega}{d\Lambda}\,dt;$$

or, expressing $d\theta$ by means of di and reducing, the second of these equations becomes

$$na^2\sqrt{(1-e^2)} = na^2\sqrt{(1-e^2)}\,\frac{\cos i}{\cos^2\Psi}\,di = \frac{1}{\cos\Psi}\,n^2a^3\frac{d\Omega}{d\Lambda}\,dt,$$

and combining this with the first of the two equations, and observing that

$$\frac{\cos^2 i}{\cos^2\Lambda\,\cos^2\Psi} + \sin^2 i = \frac{1}{\cos^2\Psi},$$

we find

$$d\,na^2\sqrt{(1-e^2)} = n^2a^3\left(\frac{\cos i}{\cos^2\Lambda}\,\frac{d\Omega}{dL} + \sin i\,\cos\Psi\,\frac{d\Omega}{d\Lambda}\right)dt,$$

$$di = \frac{na^2}{\sqrt{(1-e^2)}}\left(-\sin i\,\cos\Psi\,\frac{d\Omega}{dL} + \frac{\cos i}{\cos\Lambda\,\cos^2\Psi}\,\frac{d\Omega}{d\Lambda}\right)dt.$$

Now, considering Ω as a function of r, θ, i, Φ, then Λ, L are given as functions of θ, i, Φ by the equations $\sin\Lambda = \sin i\,\sin\Phi$, $\tan\Psi = \cos i\,\tan\Phi$, $\Psi = L - \theta$, and after some simple reductions,

$$\begin{aligned}
\frac{d\Omega}{dr} &= \frac{d\Omega}{dr}\,,\\
\frac{d\Omega}{d\theta} &= \frac{d\Omega}{dL}\,,\\
\frac{d\Omega}{di} &= \tan\Phi\left(\frac{\cos i\,\cos\Psi}{\cos\Lambda}\,\frac{d\Omega}{dL} + \cos i\,\cos\Psi\,\frac{d\Omega}{d\Lambda}\right),\\
\frac{d\Omega}{d\Phi} &= \left(\frac{\cos i}{\cos\Lambda}\,\frac{d\Omega}{dL} + \cos i\,\cos\Psi\,\frac{d\Omega}{d\Lambda}\right);
\end{aligned}$$

whence also

$$\frac{d\Omega}{d\theta} = \cos i\,\frac{d\Omega}{d\Phi} - \sin i\,\cot\Phi\,\frac{d\Omega}{di}\,.$$

We have therefore

$$\begin{aligned}
d\,na^2\sqrt{(1-e^2)} &= n^2a^3\,\frac{d\Omega}{d\Phi}\,dt,\\
di &= \frac{na\cot\Phi}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{di}\,dt,\\
d\theta &= \frac{na}{\sqrt{(1-e^2)}\,\sin i}\,\frac{d\Omega}{di}\,dt,\\
d\phi &= \frac{-na\cot i}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{di}\,dt,\\
dr &= 0,\\
d\,\frac{nae\sin f}{\sqrt{(1-e^2)}} &= n^2a^3\,\frac{d\Omega}{dr}\,dt.
\end{aligned}$$

Suppose now that we have

ρ, a radius vector, τ for t,
ϕ, a true anomaly,
u, an eccentric anomaly, &c.,

i.e. let ρ, ϕ, u, be what the radius vector, the true anomaly and the eccentric anomaly become when the time t, in so far as it enters directly, and not through the variable elements, is replaced by a new variable τ. We have

$$n\tau + c = u - e\sin u,$$

$$\phi = \tan^{-1}\frac{\sqrt{(1-e^2)}\sin u}{\cos u - e},$$

$$\rho = a\,(1-e\cos u) = \frac{a\,(1-e^2)}{1+e\cos\phi};$$

and of course the differential coefficients of ρ, ϕ with respect to τ may be at once deduced from the corresponding expressions for the differential coefficients of r, f with respect to t, the elements being considered as constant. Now, using l to denote a logarithm, and supposing that the differentiations affect only the elements, we have

$$dl\rho = \frac{da}{a} - \frac{2ede}{1-e^2} - \frac{\rho\cos\phi\,de}{a\,(1-e^2)} + \frac{\rho e\sin\phi\,d\phi}{a\,(1-e^2)},$$

$$dlr = \frac{da}{a} - \frac{2ede}{1-e^2} - \frac{r\cos f\, de}{a\,(1-e^2)} + \frac{re\sin f\, df}{a\,(1-e^2)}\,;$$

and putting for shortness

$$X_r = dlr - \frac{\rho e\sin\phi\, d\phi}{a\,(1-e^2)}\,,$$

$$X = dlr - \frac{re\sin f\, df}{a\,(1-e^2)}\,,$$

we find

$$X_r = \frac{\rho\sin\phi}{r\sin f}\,X = \left(1 - \frac{\rho\sin\phi}{r\sin f}\right)\left(\frac{da}{a} - \frac{2ede}{1-e^2}\right) - \frac{\rho}{a\,(1-e^2)}\left(\cos\phi - \frac{\sin\phi\cos r}{\sin f}\right) de,$$

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or reducing

$$X_r - \frac{\rho\sin\phi}{r\sin f}\,X = \frac{1}{\sin f\,(1+e\cos\phi)}\left\{\bigl(\sin f - \sin\phi + e\sin(f-\phi)\bigr)\left(\frac{da}{a} - \frac{2ede}{1-e^2}\right) - \sin(f-\phi)\,de\right\}.$$

Write for a moment

$$P = na^2\sqrt{(1-e^2)},\quad Q = \frac{nae\sin f}{\sqrt{(1-e^2)}}\,,\quad R = \frac{a\,(1-e^2)}{1+e\cos f}\,,$$

so that

$$a\,(1-e^2) = \frac{P^2}{n^2a^3}\,,$$

$$e^2 = \frac{1}{n^4a^6}\,P^2Q^2 + \frac{1}{n^2a^2}\,\frac{P^2}{R^2} - \frac{2}{n^2a^4}\,\frac{P^2}{R} + 1.$$

We have therefore

$$\frac{da}{a} = \frac{2ede}{1-e^2} = \frac{2dP}{P} = \frac{2}{na^2\sqrt{(1-e^2)}}\,dP,$$

$$ede = \left(\frac{1}{n^4a^6}\,P^2Q\right)dQ + \left(\frac{P^2}{n^4a^6\,R} - \frac{2}{n^2a^4}\,\frac{P^2}{R}\right)dP + \frac{P}{n^4a^6}\,\bar{Q}\,dQ + \left(-\frac{1}{n^2a^2}\,\frac{P^2}{R^2} + \frac{2}{n^2a^4}\,\frac{P^2}{R}\right)dR,$$

which, after reduction, becomes

$$de = \frac{1}{na^2\,(1-e^2)}\bigl(e\sin f + 2\,(\cos f + e\cos^2 f)\bigr)\,dP + \frac{na}{e}\,\sqrt{(1-e^2)}\,\sin f\,dQ - \frac{(1+e\cos f)^2\cos f}{a\,(1-e^2)}\,dR,$$

and substituting these values,

$$\begin{aligned} X_r - \frac{\rho\sin\phi}{r\sin f}\,X = \frac{1}{1+e\cos\phi}\Biggl\{&\bigl(2 - 2\cos(f-\phi) + e\sin f\sin(f-\phi)\bigr)\frac{1}{na^2\sqrt{(1-e^2)}}\,dP \\ &-a\,(1-e^2)\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,dQ \\ &+\frac{(1+e\cos f)^2\cos f\sin(f-\phi)}{\sqrt{(1-e^2)}}\,\frac{1}{na^2\sqrt{(1-e^2)}}\,dR\Biggr\}, \end{aligned}$$

or substituting for X, X_r, P, Q, R, their values

$$dl\rho - \frac{\rho\sin\phi}{r\sin f}\,dlr + \frac{\rho e\sin\phi}{a\,(1-e^2)}\,(df - d\phi)$$

$$= \frac{\rho}{a\,(1-e^2)}\big(2-2\cos(f-\phi)+e\sin f\sin(f-\phi)\big)\;\frac{1}{na^2\sqrt{(1-e^2)}}\;d\,na^2\sqrt{(1-e^2)}$$

$$-\rho\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\;d\,\frac{nae\sin f}{\sqrt{(1-e^2)}}$$

$$+\frac{\rho\,(1-e^2)}{r^2}\;\cot f\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,dr.$$

Now

$$-\left\{\frac{\rho}{r}\cos(f-\phi)-1+\frac{\rho}{a\,(1-e^2)}(\cos(f-\phi)-1)\right\}$$
$$=\frac{\rho}{a\,(1-e^2)}(2-2\cos(f-\phi)+e\sin f\sin(f-\phi))\,;$$

therefore

$$\begin{aligned}
&dl\rho-\frac{\rho\sin\phi}{r\sin f}\,dlr+\frac{\rho e\sin\phi}{a\,(1-e^2)}\,(df-d\phi)\\
&\quad=-\left\{\frac{\rho}{r}\cos(f-\phi)-1+\frac{\rho}{a\,(1-e^2)}(\cos(f-\phi)-1)\right\}\frac{1}{na^2\sqrt{(1-e^2)}}\,d\,na^2\sqrt{(1-e^2)}\\
&\qquad-\rho\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,d\,\frac{nae\sin f}{\sqrt{(1-e^2)}}\\
&\qquad+\frac{\rho a\,(1-e^2)}{r^2}\,\cot f\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,dr.
\end{aligned}$$

So far the variations of the elements have, in fact, been treated as independent; but if we substitute for $d\,na^2\sqrt{(1-e^2)}$, $d\,\dfrac{nae\sin f}{\sqrt{(1-e^2)}}$, dr, their values in the disturbed motion, the equation becomes

$$\begin{aligned}
dl\rho+\frac{\rho e\sin\phi}{a\,(1-e^2)}\,(df-d\phi)&=-\left\{\frac{\rho}{r}\cos(f-\phi)-1+\frac{\rho}{a\,(1-e^2)}(\cos(f-\phi)-1)\right\}\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{d\Phi}\,dt\\
&\quad-\rho\sin(f-\phi)\,\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dr}\,dt.
\end{aligned}$$

Consider now the point in which the orbit is intersected by any orthogonal trajectory to the successive positions of the orbit, or to fix the ideas, the orthogonal trajectory passing through Υ, the point in question may, for want of a recognised name, be called the "departure point;" and the angular distances in the orbit measured from this point may be termed "departures;" the expression, "the departure," is to be understood as meaning the departure of the moon. Write now

χ, the departure of the perigee,
v, the departure, $=f+\chi$,
σ, the departure of the node, $=\chi-\omega$,
Θ, the longitude in orbit of departure point, $=\theta-\sigma$.

It should be remarked that χ is not properly an element, i.e. it is not a function of a, e, c, ω, θ, i without t, and in like manner σ and Θ (which also depend upon χ) are not elements.

We have from the geometrical definition

$$d\chi = d\omega + \cos i \, d\theta;$$

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and therefore

$$d\sigma = \cos i \, d\theta,$$
$$d\Theta = (1 - \cos i) \, d\theta.$$

Moreover $v_1 = \Phi + \sigma$, which gives (assuming that the differentiations are performed with respect to t, only in so far as it enters through the variable elements) $dv_1 = d\Phi + d\sigma = d\Phi + \cos i \, d\theta$, i.e. $dv_1 = 0$, an equation which might have been assumed for the purpose of defining the departure point; the equation, in fact, expresses that the departure v_1 is measured from a point not actually fixed, but such that the increment of v_1 in the interval of time dt is the angular distance between two consecutive positions of the moon.

We have, as above noticed, $d\sigma = \cos i \, d\theta$, and thence and from what has preceded

$$di = \frac{na \cot \Phi}{\sqrt{(1 - e^2)}} \frac{d\Omega}{di} \, dt,$$
$$d\sigma = \frac{na \cot i}{\sqrt{(1 - e^2)}} \frac{d\Omega}{di} \, dt.$$

Now the position of the moon can be determined by means of the quantities r, v_1, Θ, σ, i; hence Ω (which has been considered as a function of r, Φ, θ, i) may, if we please, be considered as a function of r, v_1, Θ, σ, i and from the differential relations

$$\begin{aligned} dr &= dr, \\ dv_1 &= d\Phi + \cos i \, d\theta, \\ d\Theta &= (1 - \cos i) \, d\theta, \\ d\sigma &= d\omega + \cos i \, d\theta, \\ di &= di, \end{aligned}$$

we find

$$\begin{aligned}
\frac{d\Omega}{dr} &= \frac{d\Omega}{dr}\,, \\
\frac{d\Omega}{d\Phi} &= \frac{d\Omega}{dv_1}\,, \\
\frac{d\Omega}{d\theta} &= \cos i\left(\frac{d\Omega}{dv_1} + \frac{d\Omega}{d\sigma}\right) + (1-\cos i)\,\frac{d\Omega}{d\Theta}\,, \\
\frac{d\Omega}{di} &= \frac{d\Omega}{di}\,,
\end{aligned}$$

we have therefore

$$\frac{d\Omega}{d\theta} = \frac{1-\cos i}{\cos i}\,\frac{d\Omega}{d\Theta} = \frac{d\Omega}{d\Phi} + \frac{1}{\cos i}\,\frac{d\Omega}{d\sigma}\,,$$

or in virtue of a preceding equation

$$\frac{d\Omega}{d\sigma} + \frac{1-\cos i}{\cos i}\,\frac{d\Omega}{d\Theta} = -\tan i\,\cot\Phi\,\frac{d\Omega}{di}\,;$$

and effecting the substitutions, and collecting the results,

$$\begin{aligned}
d\,na^2\sqrt{(1-e^2)} &= n^2a^3\,\frac{d\Omega}{dv_1}\,dt,\\
dr &= 0,\\
d\,\frac{nae\sin f}{\sqrt{(1-e^2)}} &= n^2a^3\,\frac{d\Omega}{dr}\,dt,\\
di &= \frac{na\cot i}{\sqrt{(1-e^2)}}\left(\frac{d\Omega}{d\sigma}+\frac{1-\cos i}{\cos i}\,\frac{d\Omega}{d\Theta}\right)dt,\\
d\sigma &= \frac{na\cot i}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{di}\,dt,
\end{aligned}$$

where Ω is considered as a function of r, v_1, Θ, σ, i.

Instead of σ, i we may introduce the new quantities p, q defined by the equations

$$p=\sin i\,\sin\sigma,\quad q=\sin i\,\cos\sigma;$$

this gives $\sin^2 i=p^2+q^2$, $\sigma=\tan^{-1}\dfrac{p}{q}$ and retaining in the formulæ the sine and cosine of i, to avoid the introduction of irrational functions of p^2+q^2, we have

$$d\Theta=(1-\cos i)\,d\theta=\frac{1-\cos i}{\cos i}\,d\sigma=\frac{1-\cos i}{\sin i\,\sin^2 i}\,(qdp-pdq),$$

i.e.

$$d\Theta=\frac{qdp-pdq}{\cos i\,(1+\cos i)}\,,$$

which determines Θ by means of p and q. We have moreover

$$\begin{aligned}
dp &= \sin i\,\cos\sigma\,d\sigma+\cos i\,\sin\sigma\,di,\\
dq &= -\sin i\,\sin\sigma\,d\sigma+\cos i\,\cos\sigma\,di,\\
\frac{d\Omega}{dp} &= \frac{\sin\sigma}{\cos i}\,\frac{d\Omega}{di}+\frac{\cos\sigma}{\sin i}\,\frac{d\Omega}{d\sigma}\,,\\
\frac{d\Omega}{dq} &= \frac{\cos\sigma}{\cos i}\,\frac{d\Omega}{di}-\frac{\sin\sigma}{\sin i}\,\frac{d\Omega}{d\sigma}\,,
\end{aligned}$$

from which equations and the foregoing values of di and $d\sigma$ we find the values of dp and dq; the other equations of the system remain unaltered, and we have therefore

$$d\,na^2\sqrt{(1-e^2)} = n^2a^3\,\frac{d\Omega}{dv_1}\,dt,$$

$$dr = 0,$$

$$d\,\frac{nae\sin f}{\sqrt{(1-e^2)}} = n^2a^3\,\frac{d\Omega}{dr}\,dt,$$

$$dp = \frac{na\cos^2 i}{\sqrt{(1-e^2)}}\left(\frac{d\Omega}{dq} - \frac{p}{\cos i\,(1+\cos i)}\,\frac{d\Omega}{d\Theta}\right)dt,$$
$$dq = -\frac{na\cos^2 i}{\sqrt{(1-e^2)}}\left(\frac{d\Omega}{dp} + \frac{q}{\cos i\,(1+\cos i)}\,\frac{d\Omega}{d\Theta}\right)dt,$$

where Ω is considered as a function of r, v_1, Θ, p, q. The symbols $\dfrac{d\Omega}{dp}$, $\dfrac{d\Omega}{dq}$, as employed by Hansen, mean that the differentiations are to be performed as if Θ was a function of p, q, such that

$$d\Theta = \frac{d\Theta}{dp}\,dp + \frac{d\Theta}{dq}\,dq = \frac{qdp - pdq}{\cos i\,(1+\cos i)}\,;$$

the last two equations being therefore nothing else than what Hansen represents by

$$dp = \frac{na\cos^2 i}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dq}\,dt,$$
$$dq = -\frac{na\cos^2 i}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dp}\,dt.$$

Write now

λ, the departure, τ for t,

i.e. λ is what v_1 becomes when t, in so far as it enters explicitly, and not through the variable elements, is replaced by the new variable τ; so that, in fact $\lambda = \phi + \chi$, The values of r, v_1 could be at once found from those of ρ, λ by changing τ into t; and to determine the values of ρ, λ, Hansen proceeds as follows: writing

$$\lambda = \Pi\,(\zeta, t),$$
$$l\rho = \Gamma\,(\zeta, t) + \beta,$$

where ζ and β are new variables functions of τ and t, Π, Γ are arbitrary functional symbols; so that if z, w are what ζ, β become when τ is changed into t, we should have

$$v_1 = \Pi\,(z, t),$$
$$lr = \Gamma\,(z, t) + w;$$

then the foregoing equations give

$$\begin{aligned}
\frac{d\lambda}{d\tau} &= \Pi'(\zeta, t)\,\frac{d\zeta}{d\tau}\,, \\
\frac{d\lambda}{dt} &= \Pi'(\zeta, t)\,\frac{d\zeta}{dt} + \Pi_{,}(\zeta, t)\,, \\
\frac{dl\rho}{d\tau} &= \Gamma'(\zeta, t)\,\frac{dl\rho}{d\tau} + \frac{d\beta}{d\tau}\,, \\
\frac{dl\rho}{dt} &= \Gamma''(\zeta, t)\,\frac{d\zeta}{dt} + \Gamma_{,}(\zeta, t) + \frac{d\beta}{dt}\,;
\end{aligned}$$

[where the accents and strokes denote differentiation in regard to ζ, t respectively].

Hence eliminating $\Pi'(\zeta, t)$ and $\Gamma'(\zeta, t)$ we have

$$\frac{d\lambda}{d\tau}\frac{d\zeta}{dt} - \frac{d\lambda}{dt}\frac{d\zeta}{d\tau} = -\Pi_{,}(\zeta, t)\frac{d\zeta}{d\tau},$$

$$\frac{dl\rho}{d\tau}\frac{d\beta}{dt} - \frac{dl\rho}{dt}\frac{d\beta}{d\tau} = \frac{d\beta}{d\zeta}\frac{d\zeta}{dt}\frac{d\beta}{d\tau} + \frac{d\beta}{d\zeta}\frac{d\zeta}{d\tau} - \Gamma_{,}(\zeta, t)\frac{d\zeta}{d\tau},$$

or, what is the same thing,

$$\frac{\frac{d\zeta}{dt}}{\frac{d\zeta}{d\tau}} - \frac{d\lambda}{dt} = -\Pi_{,}(\zeta, t)\frac{1}{\frac{d\zeta}{d\tau}},$$

$$\frac{d\beta}{dt} - \frac{\frac{d\zeta}{dt}}{\frac{d\zeta}{d\tau}}\frac{dl\rho}{d\tau} = \frac{dl\rho}{dt} - \frac{dl\rho}{d\tau}\frac{\frac{d\zeta}{dt}}{\frac{d\zeta}{d\tau}} + \Pi_{,}(\zeta, t)\frac{dl\rho}{d\zeta}\frac{d\zeta}{d\tau} - \Gamma_{,}(\zeta, t),$$

or writing

$$T = \frac{d\lambda}{dt} - \frac{\frac{d\zeta}{dt}}{\frac{d\zeta}{d\tau}}, \quad R = \frac{dl\rho}{dt} - \frac{d\beta}{dt}\frac{\frac{d\zeta}{dt}}{\frac{d\zeta}{d\tau}},$$

we have

$$T = \frac{1}{\frac{d\zeta}{d\tau}}\left\{\frac{d\lambda}{d\tau}\left(\frac{d\zeta}{dt}\right)\frac{d\lambda}{dt} + \Pi_{,}(\zeta, t)\left(\frac{d\zeta}{d\tau}\right)\right\} - \Pi_{,}(\zeta, t)\frac{\frac{d\zeta}{dt}}{\frac{d\zeta}{d\tau}},$$

$$R = \frac{dl\rho}{dt} - \frac{\frac{d\zeta}{dt}}{\frac{d\zeta}{d\tau}}\frac{d\lambda}{d\tau} + \Pi_{,}(\zeta, t)\frac{dl\rho}{d\zeta}\frac{d\zeta}{d\tau} - \Gamma_{,}(\zeta, t).$$

Now from the equation $\lambda = \phi + \chi$ where χ is independent of τ, the differential coefficients of λ and ρ with respect to τ are at once deduced from those of f, r with respect to t, and we have,

$$\frac{d^2\lambda}{d\tau^2} = -\frac{2n^2a^4}{\rho^3}e\sin\phi,$$

$$\frac{d\lambda}{d\tau} = \frac{na^2\sqrt{(1-e^2)}}{\rho^2},$$

$$\frac{dl\rho}{d\tau} = \frac{nae\sin\phi}{\rho\sqrt{(1-e^2)}}.$$

Consequently

$$T = \frac{\rho^2}{na^2\sqrt{(1-e^2)}}\left\{\frac{d}{dt}\,\frac{na^2\sqrt{(1-e^2)}}{\rho^2} + \frac{2\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt}\right.$$

$$\left. -\frac{2\rho e\sin\phi}{a\,(1-e^2)}\,\Pi_,(\zeta,t) - \frac{\rho^2}{na^2\sqrt{(1-e^2)}}\,\Pi'_,(\zeta,t)\,\frac{d\zeta}{dt}\right\}$$

$$= -2\,\frac{dl\rho}{dt} + \frac{2\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt} + \frac{1}{na^2\sqrt{(1-e^2)}}\,\frac{d}{dt}\,na^2\sqrt{(1-e^2)}$$

$$-\frac{2\rho e\sin\phi}{a\,(1-e^2)}\,\Pi_,(\zeta,t) - \frac{\rho^2}{na^2\sqrt{(1-e^2)}}\,\Pi'_,(\zeta,t)\,\frac{d\zeta}{dt}\,,$$

$$R = \frac{dl\rho}{dt} - \frac{\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt} + \frac{\rho e\sin\phi}{a\,(1-e^2)}\,\Pi_,(\zeta,t) - \Gamma_,(\zeta,t),$$

and substituting in these equations the values of

$$\frac{dl\rho}{dt} = \frac{\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt} \quad \text{and} \quad d\,na^2\sqrt{(1-e^2)}$$

we find

$$T = \left\{2\,\frac{\rho}{r}\cos(v_1-\lambda) - 1 + \frac{2\rho}{a\,(1-e^2)}(\cos(v_1-\lambda)-1)\right\}\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dv_1} + 2\,\frac{\rho}{r}\sin(v_1-\lambda)\,\frac{na}{\sqrt{(1-e^2)}}\,r$$

$$-2\Pi_,(\zeta,t) + \frac{\rho e\sin\phi}{a\,(1-e^2)}\,\Pi_,(\zeta,t)\,\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dv_1} - 2\Pi_,(\zeta,t) - \frac{\rho^2}{na^2\sqrt{(1-e^2)}}\,\Pi'_,(\zeta,t)\,\frac{d\zeta}{dt}\,,$$

$$R = -\left\{\frac{\rho}{r}\cos(v_1-\lambda) - 1 + \frac{\rho}{a\,(1-e^2)}(\cos(v_1-\lambda)-1)\right\}\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dv_1} - \frac{\rho}{r}\sin(v_1-\lambda)\,\frac{na}{\sqrt{(1-e^2)}}\,r$$

$$+\,\Pi_,(\zeta,t)\,\frac{\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt} - \Gamma_,(\zeta,t),$$

which are Hansen's values, except that $\dfrac{d\lambda}{d\tau}$ in the coefficient of $\Pi_,(\zeta,t)$ has been replaced by its value.

2, Stone Buildings, 31st March, 1855.

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On Gauss' Method for the Attraction of Ellipsoids.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 162–166.]

The following is the method employed in Gauss' Memoir "Theoria Attractionis Corporum Sphæroidicorum ellipticorum homogæneorum methodo novo tractata," 1813. *Gauss. Gott. recent.*, t. II. [and *Werke* t. V. pp. 1–22]. I have somewhat developed the geometrical considerations upon which the method depends.

The attraction of the ellipsoid is found by means of the following theorems, which apply generally to the case of a homogeneous solid bounded by a closed surface:— M denotes the attracted point, P a point of the surface, PQ is the normal (lying outside the surface) at the point P, dS is the element of the surface at this point, MQ, QX, and MX denote angles at the point P, viz. MQ the $\angle MPQ$, and QX and MX the inclinations of QP and MP respectively to a line PX drawn in a direction assumed as that of the axis of X, MP denotes the distance between the points M and P. And X is the attraction in the direction opposite to that of the axis of x; the integrations extend over the entire surface.

Theorem. The integral

$$\iint \frac{dS \cos MQ}{MP^n}$$

has for its value

$$0, \quad -2\pi, \quad \text{or} \quad -4\pi,$$

according as M is exterior to, upon, or interior to, the surface.

This is obviously a purely geometrical theorem.

Theorem. The attraction is given by the formula

$$X = \iint \frac{dS \cos QX}{MP}.$$

Theorem. The attraction is also given by the formula

$$X = -\iint \frac{dS \cos MQ \cos MX}{MP^2}.$$

Consider now an ellipsoid, the semi-axes of which are A, B, C, and putting a, b, c for the coordinates of the attracted point M, and x, y, z for the coordinates of P, assume

$$x = A\xi,$$
$$y = B\eta,$$
$$z = C\zeta,$$

so that ξ, η, ζ are the coordinates of a point P' on a sphere, radius unity, corresponding in a definite manner to the point P on the ellipsoid; and let $d\sigma$ be the corresponding element of the spherical surface, we have

$$dS = \frac{ABC}{p}\, d\sigma,$$

where p denotes the perpendicular let fall from the centre of the ellipsoid upon the tangent plane at P.

Moreover,

$$\cos QX = \frac{p\xi}{A}, \quad \cos MX = \frac{a - x}{MP} = \frac{a - A\xi}{MP};$$

and therefore

$$dS \cos QX = \frac{BC\xi\, d\sigma}{MP}.$$

The second theorem gives therefore

$$A\,\frac{X}{ABC} = \iint \frac{\xi\, d\sigma}{MP},$$

where the integration is extended over the surface of the sphere, and the third theorem gives

$$\frac{X}{ABC} = -\iint \frac{(a - A\xi)\cos MQ\, dS}{MP^3},$$

where the integration is extended over the surface of the ellipsoid.

Suppose now a confocal ellipsoid, the semi-axes of which are $A+\delta A$, $B+\delta B$, $C+\delta C$, and let P' be the point on this new ellipsoid which *corresponds* to the point P on the original ellipsoid, i.e. let P' be the point whose coordinates are $(A+\delta A)\xi$, $(B+\delta B)\eta$, $(C+\delta C)\zeta$; the decrement of MP will be equal to the normal distance δN between the two ellipsoids at the point P, multiplied into the cosine of the angle MQ, and we have, by a property of confocal ellipsoids, $A\delta A = B\delta B = C\delta C = p\delta N$; we have therefore

$$\delta\,\overline{MP} = -\frac{A\delta A\cos MQ}{p}\,.$$

which gives

$$d\sigma\,.\,\delta MP = \frac{A\delta A}{ABC}\,dS\cos MQ.$$

Now from the equation

$$A\,\frac{X}{ABC} = \iint \frac{\xi\,d\sigma}{MP}\,,$$

we find

$$A\delta\,\frac{X}{ABC} + \frac{X}{ABC}\,\delta A = \iint \frac{\xi\,d\sigma\,\delta\overline{MP}}{MP^2} = -\frac{\delta A}{ABC}\iint \frac{A\xi\cos MQ\,dS}{MP^2}\,.$$

But

$$-\frac{X}{ABC}\,\delta A = \frac{\delta A}{ABC}\iint \frac{(a - A\xi)\cos MQ\,dS}{MP^2}\,,$$

and consequently

$$A\delta\,\frac{X}{ABC} = \frac{a\delta A}{ABC}\iint \frac{\cos MQ\,dS}{MP^2}\,.$$

Hence, by the first theorem,

In the case of an exterior point, we have

$$\delta\,\frac{X}{ABC} = 0,$$

i.e. the attractions, in the directions of the axes, of confocal ellipsoids vary as the masses; which is Maclaurin's theorem for the attractions of ellipsoids upon an exterior point.

In the case of an interior point, we have

$$\delta\,\frac{X}{ABC} = -4\pi a\,\frac{a\delta A}{A^2BC}\,,$$

or, taking α, β, γ as the semi-axes of an ellipsoid confocal with the ellipsoid (A, B, C), but exterior to it, and supposing that (X) refers to the ellipsoid (α, β, γ), we have

$$\delta\left(\frac{(X)}{\alpha\beta\gamma}\right) = -4\pi a\,\frac{\delta\alpha}{\alpha^2\beta\gamma}\,.$$

Now introducing instead of α the new variable θ, such that $\alpha^2 = A^2 + \theta$, we have

$\dfrac{\delta\alpha}{\alpha^2} = \dfrac{\delta\theta}{(A^2+\theta)^{\frac{3}{2}}}$, $\quad \beta = (B+\theta)^{\frac{1}{2}}$, $\gamma = (C^2+\theta)^{\frac{1}{2}}$, and consequently writing d for δ,

$$d\left(\frac{(X)}{\alpha\beta\gamma}\right) = -4\pi a \,\frac{d\theta}{(A^2+\theta)^{\frac{3}{2}}(B^2+\theta)^{\frac{1}{2}}(C^2+\theta)^{\frac{1}{2}}}\,.$$

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and thence, effecting the integration,

$$\frac{X_{\prime}}{(A^2+\theta_{\prime})^{\frac{1}{2}}(B+\theta_{\prime})^{\frac{1}{2}}(C^2+\theta_{\prime})^{\frac{1}{2}}}-\frac{X}{ABC}=-4\pi a\int_{\theta}^{\theta_{\prime}}\frac{d\theta}{(A^2+\theta)^{\frac{3}{2}}(B^2+\theta)^{\frac{1}{2}}(C^2+\theta)^{\frac{1}{2}}},$$

where $X_{\prime}$ refers to the ellipsoid whose semi-axes are $(A^2+\theta_{\prime})^{\frac{1}{2}}$, $(B^2+\theta_{\prime})^{\frac{1}{2}}$, $(C^2+\theta_{\prime})^{\frac{1}{2}}$. In the case where $\theta_{\prime}=\infty$, we have

$$\frac{X_{\prime}}{(A^2+\theta_{\prime})^{\frac{1}{2}}(B+\theta_{\prime})^{\frac{1}{2}}(C^2+\theta_{\prime})^{\frac{1}{2}}}=0,$$

and consequently

$$X=4\pi a\,ABC\int_0^{\infty}\frac{d\theta}{(A^2+\theta)^{\frac{3}{2}}(B^2+\theta)^{\frac{1}{2}}(C^2+\theta)^{\frac{1}{2}}},$$

which is the expression for the attraction, in the direction opposite to that of the axis of x, of the ellipsoid (A,B,C) upon an interior point, the coordinates of which are (a,b,c).

In the case of an exterior point, let $A_{\prime}$, $B_{\prime}$, $C_{\prime}$ be the semi-axes of the confocal ellipsoid passing through the attracted point; so that putting $A_{\prime}=\sqrt{(A^2+\eta)}$, $B_{\prime}=\sqrt{(B^2+\eta)}$, $C_{\prime}=\sqrt{(C^2+\eta)}$, we have

$$\frac{a^2}{A^2+\eta}+\frac{b^2}{B^2+\eta}+\frac{c^2}{C^2+\eta}=1,$$

the attraction is equal to $\dfrac{ABC}{A_{\prime}B_{\prime}C_{\prime}}\times$ attraction of the ellipsoid which passes through the point, i.e.

$$X=4\pi a\,ABC\int_{\eta}^{\infty}\frac{d\theta}{(A_{\prime}^2+\theta)^{\frac{3}{2}}(B_{\prime}^2+\theta)^{\frac{1}{2}}(C_{\prime}^2+\theta)^{\frac{1}{2}}},$$

or, putting $\theta-\eta$ instead of θ,

$$X=4\pi a\,ABC\int_{\eta}^{\infty}\frac{d\theta}{(A^2+\theta)^{\frac{3}{2}}(B^2+\theta)^{\frac{1}{2}}(C^2+\theta)^{\frac{1}{2}}},$$

which is the expression for the attraction upon an exterior point. The formulæ coincide, as they ought to do, in the case of a point upon the surface.

2, Stone Buildings, 9th April, 1855.